

Zap! Analyzing Lightning and Satellite Imagery of Tropical Cyclones for Pattern Recognition of Intensity Changes



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Summary

We analyzed lightning data for 43 hurricanes between 2019 and 2023, investigating the relationship between lightning activity and changes in hurricane intensity. We also created an interactive app to enable others to easily explore and query this data.

Background & Motivation

Lightning is a proxy for deep convection, the transfer of heat energy from the warm ocean to the upper atmosphere that powers hurricanes. Previous research has found associations between temporal and spatial patterns of lightning and tropical cyclone intensity changes.

In recent years, high-quality lightning data has become available from the Geostationary Lightning Mapper (GLM), an optical satellite sensor which can detect an estimated 70-90% of lightning activity across the Americas and adjacent ocean regions.

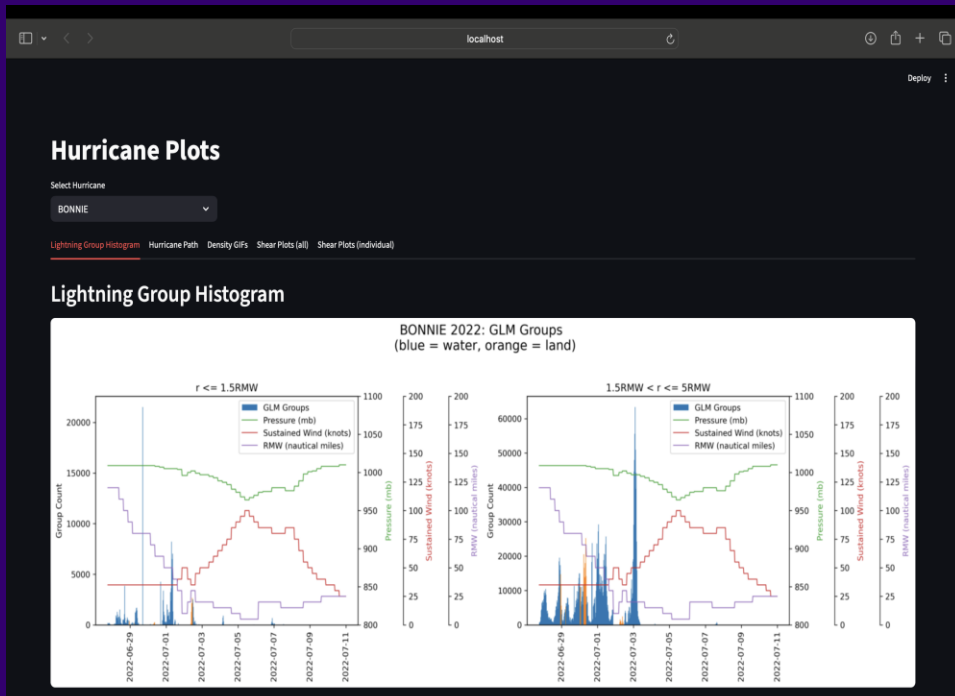
Forecasters have traditionally used cloud patterns to assess and predict hurricane intensity change, a method called the Dvorak Technique. While widely used, this method is imprecise and subject to forecaster bias. We aimed to identify patterns in the GLM data which could provide additional forecasting insights.

Data Sources

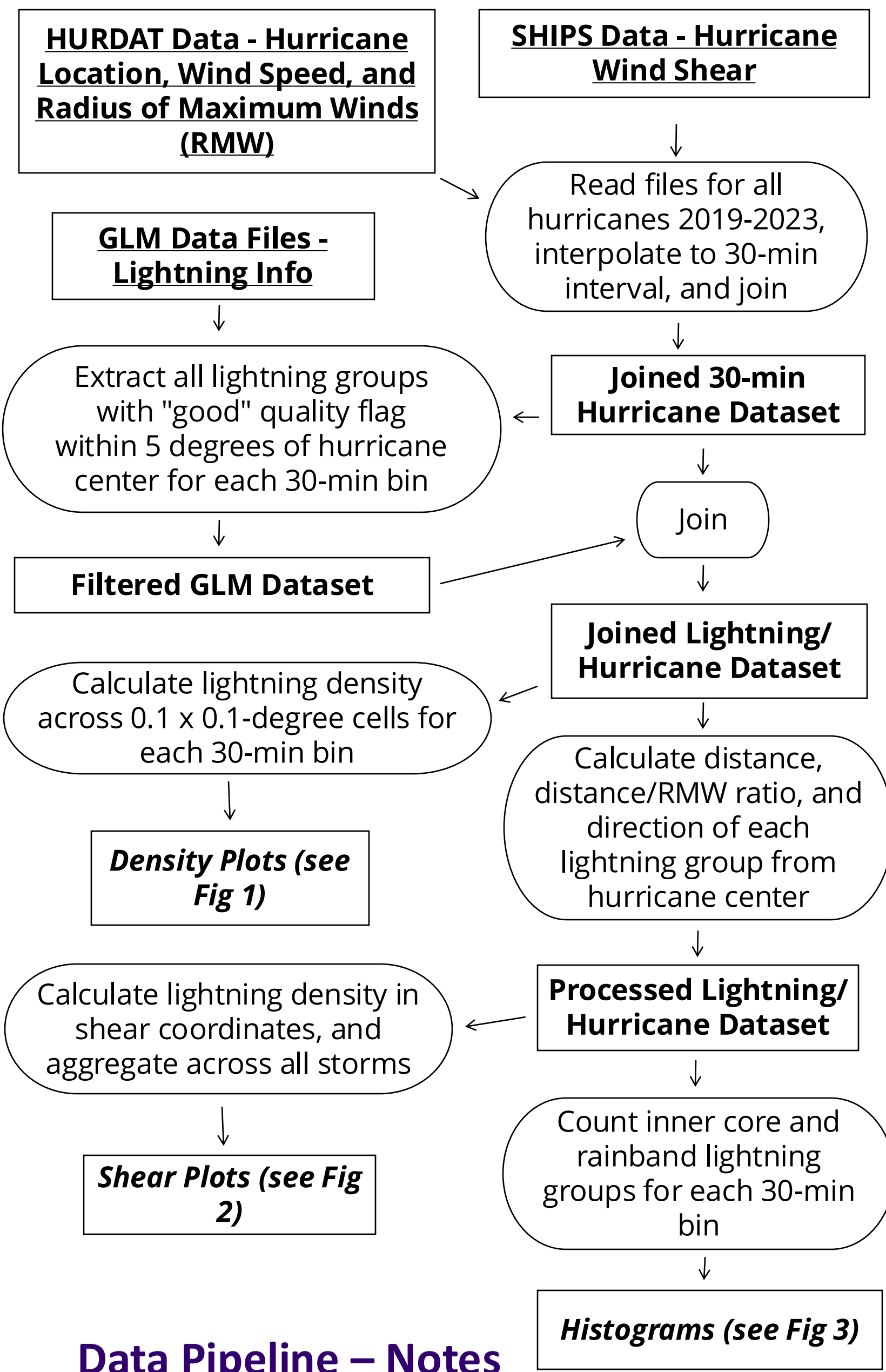
- NASA GOES-R Satellite Geostationary Lightning Mapper Sensors (GLM): (NetCDF)
 - Lightning groups, with timestamp, position, and quality flag
- NOAA Storm Tracker (HURDAT): (txt)
 - Storm center coordinates, wind speed, and pressure at 6-hour interval
- Colorado State – Statistical Hurricane Intensity Prediction Scheme (SHIPS): (txt)
 - Vertical wind shear magnitude (850-200mb) averaged across storm vicinity, at 6-hour interval

Results Available in Local Interactive Streamlit Website!

View all analysis plots for any tropical cyclone between 2019 and 2023



Data Pipeline – Simplified Diagram



Data Pipeline – Notes

- There was one GLM file every 20 seconds. This added up, and it was a big challenge to write code that processed all those files reasonably quickly – lots of multithreading!
- Wind shear is the difference in wind speed and direction between the lower and upper atmosphere. It greatly impacts lightning distribution in hurricanes, so many previous studies use a shear coordinate system.
- We also calculated wind speed change to categorize storms as rapidly weakening (RW), weakening (W), no change (NC), intensifying (I), or rapidly intensifying (RI) at each timestamp. We then enabled our app to query data by this intensification status (see Fig 2).

Lightning Analysis Visualizations

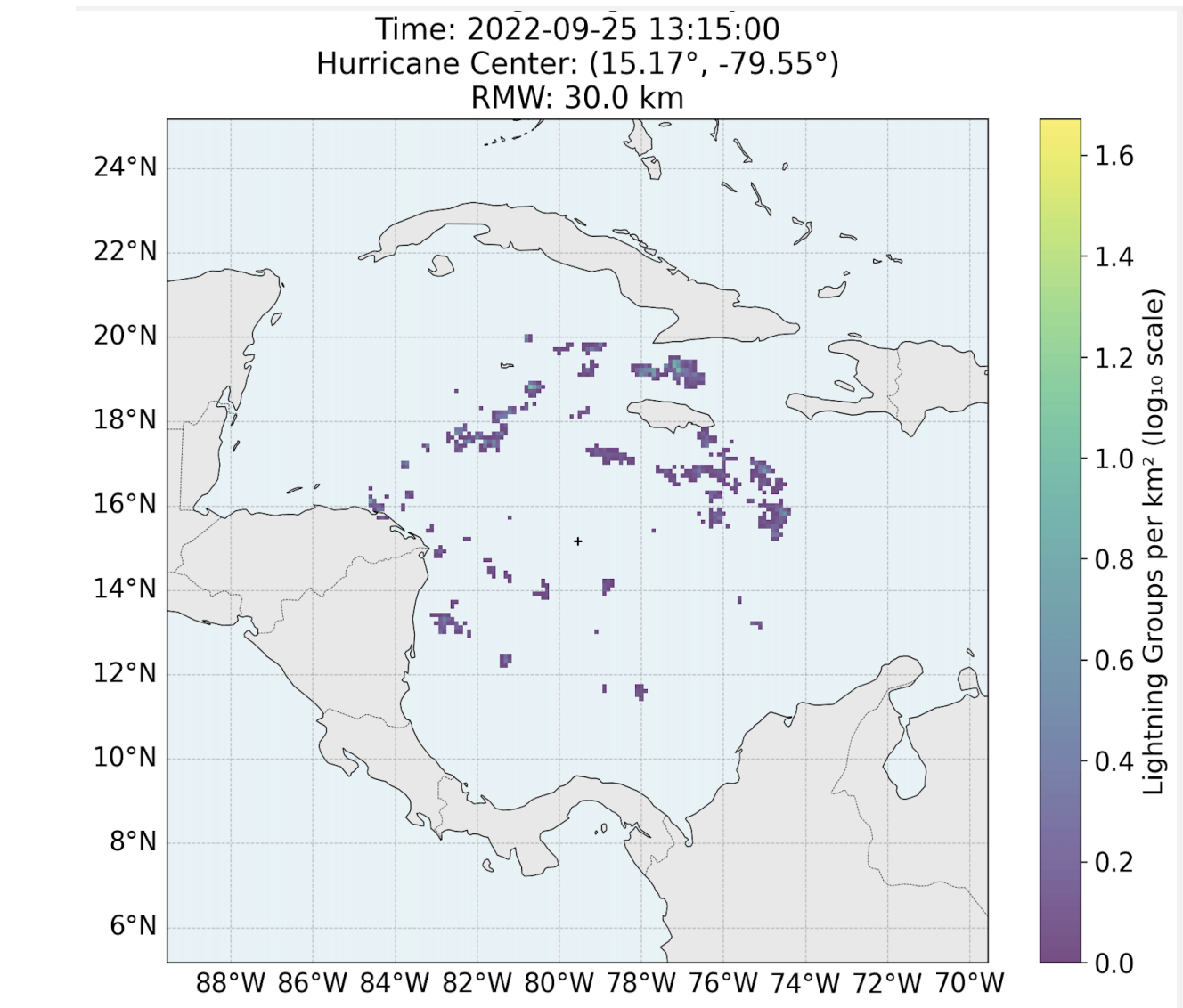


Figure 1. Lightning Density for Hurricane Ian

The lightning density plot displays individual time bins as frames in an animated GIF, centered on the storm's best track. Lightning observations are divided into $0.1^\circ \times 0.1^\circ$ grid cells, where density is calculated as the lightning group count per cell divided by the cell area and displayed on a logarithmic scale.

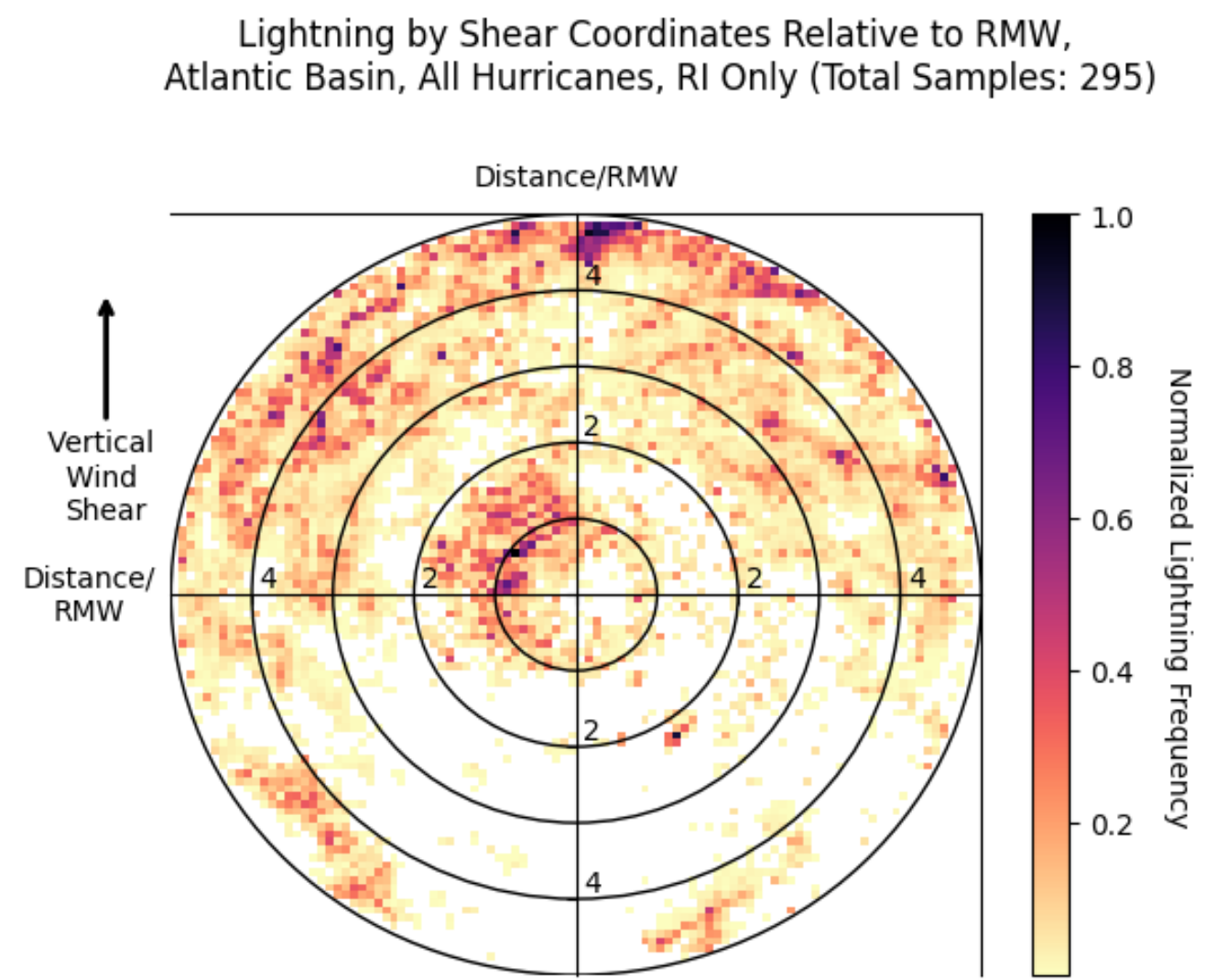


Figure 2. Average lightning density for rapidly intensifying hurricanes

This plot displays the average lightning density aggregated across all rapidly intensifying hurricanes – that is, hurricanes whose wind speed increases by more than 30kt in the following 24 hours. Locations of lightning relative to the hurricane center are displayed in shear coordinates and scaled to the hurricane RMW.

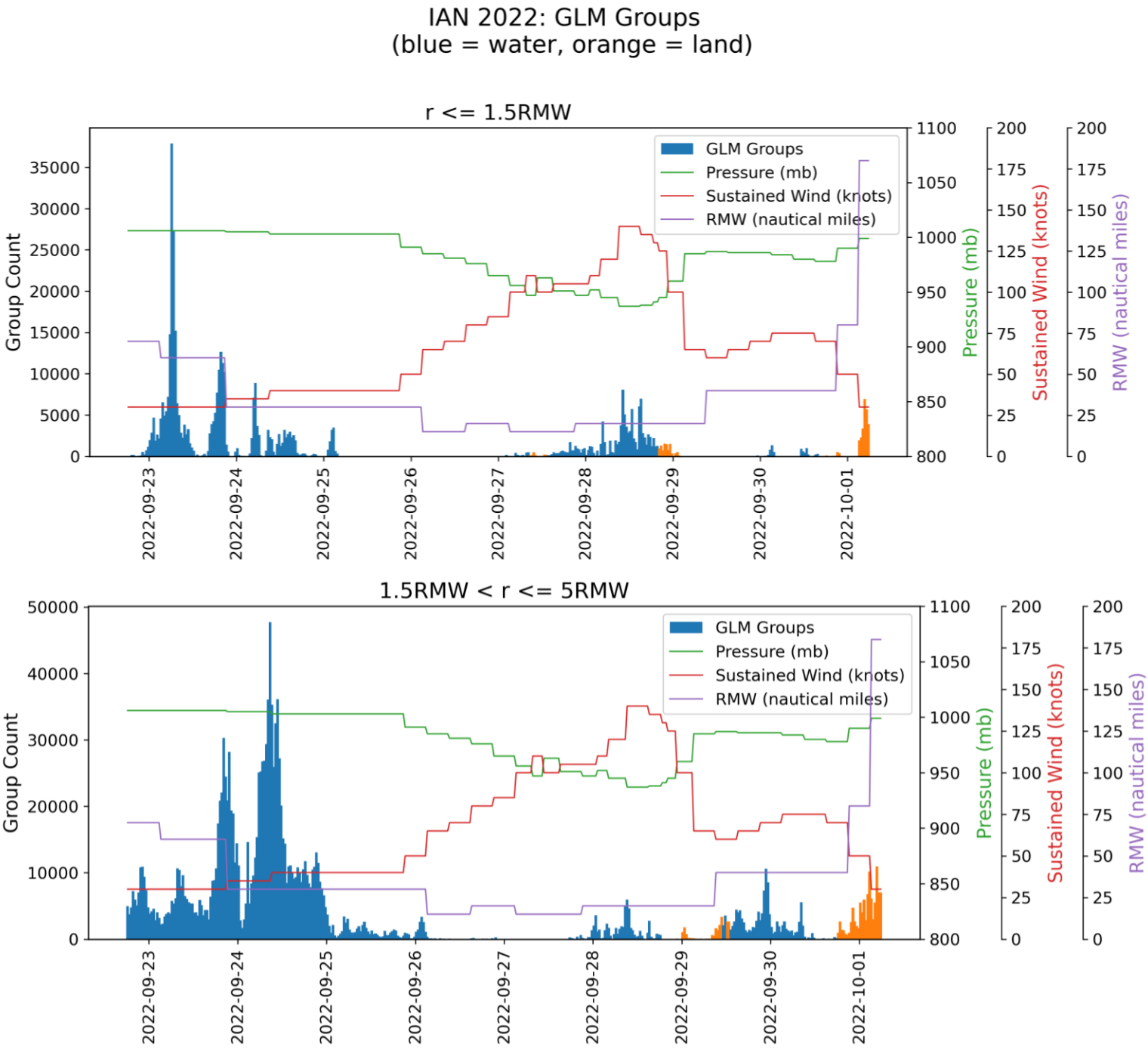


Figure 3. GLM Lightning Groups for Hurricane Ian

These plots contain 30min counts of GLM lightning groups for hurricane Ian along with data on minimum pressure, maximum sustained winds, and the radius of maximum winds for the storm. The first plot shows lightning groups in the inner core (within 1.5RMW from the storm center) and the second plot shows groups in the rainbands (between 1.5RMW and 5RMW from the center). We see that Ian reached minimum pressure and maximum sustained winds on 9/28. We also see that there is a spike in lightning activity on that day, especially in the inner core.

Future Work

Use machine learning techniques to model hurricane intensity and intensity changes using lightning data and infrared images.

References

Dvorak, V. F., 1972: A technique for the analysis and forecasting of tropical cyclone intensities from satellite pictures. NOAA Tech. Memo. NESS 36, 15 pp.

Instruments: Geostationary Lightning Mapper (GLM) (n.d.). Geostationary Operational Environmental Satellites - R Series; NASA. Retrieved December 11, 2025, from <https://www.goes-r.gov/spacesegment/glm.html>.

Lin, S.-J.; Chou, K.-H. The lightning distribution of tropical cyclones over the Western North Pacific. Mon. Weather Rev. 2020, 148, 4415–4434.

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